2.3.1 Perimeter of combined shapes

The perimeter of a combined shape is the sum of all the lengths of the sides that form a closed boundary.

For example, observe the Fig. 2.19 which has a square of side *a units* and an equilateral triangle of side *a units* combined together.

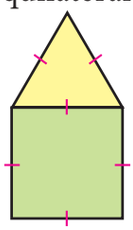


Fig. 2.19

Though a square has 4 sides and an equilateral triangle has 3 sides, when they are combined together, we will have a total of only 5 sides as the boundary of the combined shape and not 7 sides. Isn't it? So, the perimeter of the combined shape, here is 5*a units*.

2.3.2 Area of combined shapes

To find the area of a combined shape, split the combined shape into known simpler shapes, find their area separately and then add them up. That is, the area of combined shapes is nothing but the sum of all the areas of the simple shapes in it.

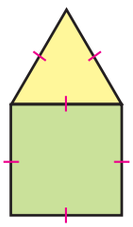


Fig. 2.20

To find the area of the Fig.2.20, find the area of the square and the area of the equilateral triangle separately and then add them up.

The combined shapes that we come across mostly in our day-to-day life are irregular polygons. To find the area of the given irregular polygon, we must identify the simple shapes in it, find their areas separately and then add them up.

For example, an irregular polygonal field given below can be split into known simpler shapes as under and then its area can be found.

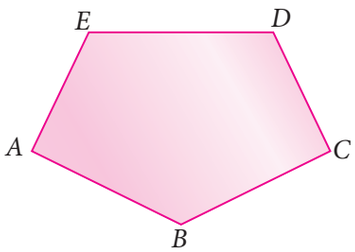


Fig. 2.21

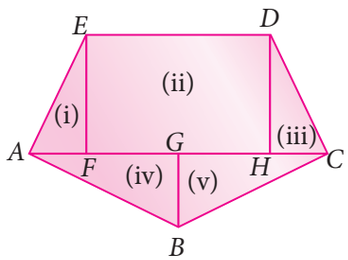


Fig. 2.22

DO YOU KNOW?

A closed plane figure formed by three or more sides is called a 'polygon' .Based on the sides, some of the polygons are named as given below.

Number of sides	3	4	5	6	7	8	9	10
Name of the Polygon	Triangle	Quadrilateral	Pentagon	Hexagon	Heptagon	Octagon	Nonagon	Decagon

If all sides and all angles of a polygon are equal, then it is called as a regular polygon. Examples: equilateral triangle, square etc., Other polygons are irregular polygons. Examples: scalene triangle, rectangle etc.,

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